END-TO-END RESEARCH

PRIVACY-PRESERVING FEDERATED LEARNING

Brain Tumor MRI Classification

A multi-source experimental framework for distributed image learning, privacy–utility analysis and explainable AI

SOURCE / CLIENT 01

SOURCE / CLIENT 02

SOURCE / CLIENT 03

Four classes • Three simulated source cohorts • CNN + Federated Optimisation

Research / education prototype — not a clinical diagnostic device

Why federated learning for brain MRI?

The notebook studies the balance between predictive utility, decentralised data ownership and system-level cost.

THE DATA CHALLENGE

MRI image collections may be held by separate organisations. Moving all raw images into one central pool can be difficult under privacy, governance and data-sharing constraints.

THE RESEARCH IDEA

Treat three independently sourced Kaggle cohorts as simulated hospital clients. Train locally, exchange model parameters and aggregate a shared global model.

THE QUESTION

How do data heterogeneity, aggregation strategy, update privacy and communication compression affect model quality and client-level outcomes?

FOUR CLASS LABELS

GLIOMA | MENINGIOMA | PITUITARY | NO TUMOR

Scope note: the notebook's 'hospitals' are simulated cohorts—not verified hospitals. Source overlap and lack of patient identifiers are documented limitations.

Three source cohorts become three local clients

The notebook downloads each source independently, inspects its folder structure and builds a harmonised image index.

COHORT 01

masoudnickparvar /
brain-tumor-mri-dataset

Independent KaggleHub download

COHORT 02

sartajbhuvaji /
brain-tumor-classification-mri

Independent KaggleHub download

COHORT 03

mohiburrahmanrifat /
brain-tumor-mri

Independent KaggleHub download

AUTHENTICATE

Secure kaggle.json upload

INSPECT

Folder tree + image files

HARMONISE

Map aliases to 4 labels

DEDUPLICATE

Perceptual-hash filtering

SPLIT

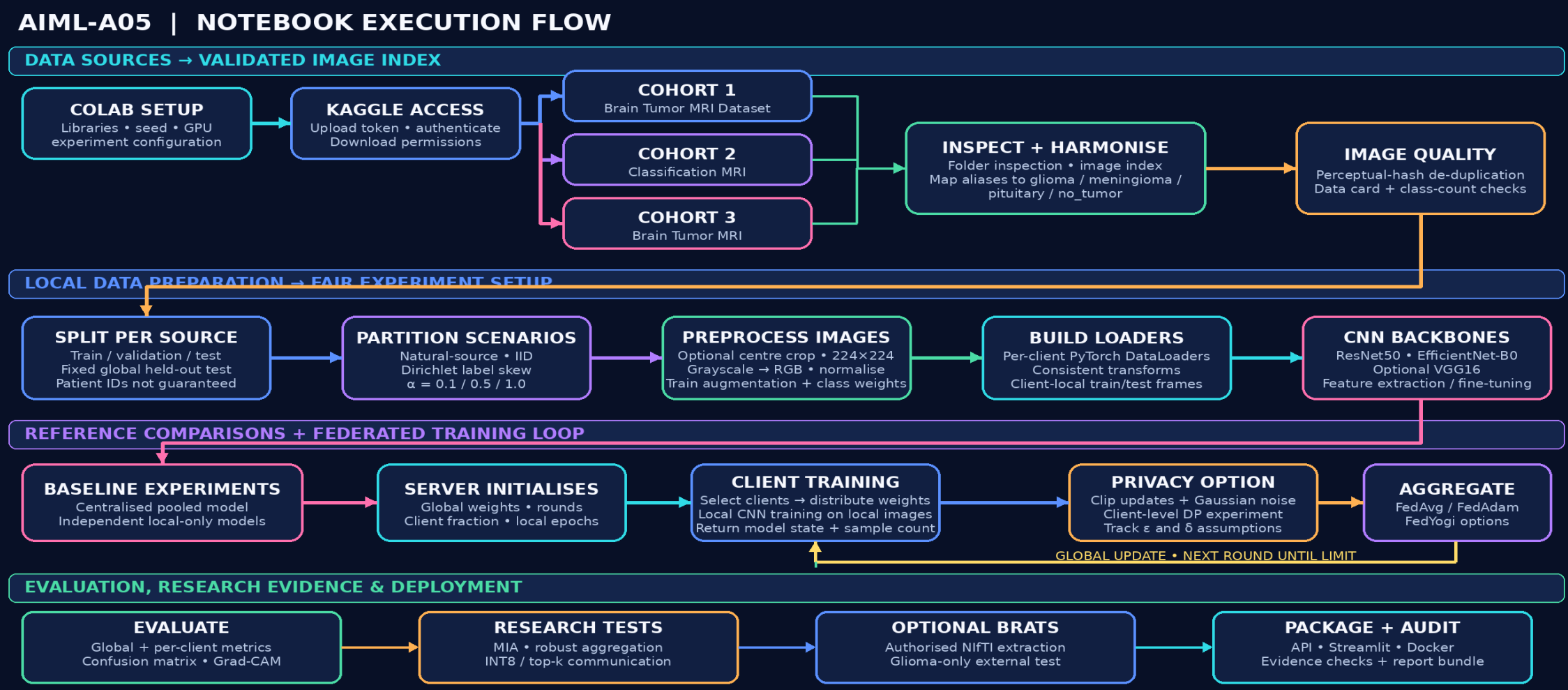
Local train / val / test

DATA GOVERNANCE CHECKS

Review class counts • record dataset slugs and access terms • disclose source overlap • image-level pHash ≠ patient-level de-duplication • patient IDs are not guaranteed.

Actual notebook execution flow

A process-oriented flowchart: data enters, decisions and loops occur, results are tested, then the project is packaged.



From MRI image files to comparable CNN experiments

The same four-class input pipeline supports independent, pooled and federated comparisons.

IMAGE STANDARDISATION

Optional centre crop
Resize to 224 × 224
Grayscale → 3-channel RGB
Per-image intensity normalisation

TRAINING PIPELINE

Training-only augmentation
Class-weight calculation
Weighted sampling option
Per-client PyTorch DataLoaders

MODEL BACKBONES

ResNet50
EfficientNet-B0
Optional VGG16
Feature extraction or fine-tuning

THREE REFERENCE VIEWS

CENTRALISED

Pooled training data; upper-bound reference

LOCAL-ONLY

One independent model per source/client

FEDERATED

Shared model trained through rounds

Fair comparison requires matched model settings, optimiser, data budget, augmentation, epochs and seeds.

Inside one federated training round

The manual PyTorch simulation keeps source images in client DataLoaders and returns model states to the coordinator.

1. **SERVER** — Initial global weights
2. **SELECT** — Choose participating clients
3. **DISTRIBUTE** — Send current parameters
4. **TRAIN LOCALLY** — Train CNN on local images
5. **RETURN** — Model state + sample count
6. **AGGREGATE** — Weighted FedAvg / adaptive
7. **UPDATE** — Refresh global model

FedAvg

Sample-count weighted model averaging

FedProx

Local objective adds proximal penalty

FedBN

Keep client-specific BatchNorm state

FedAdam / Yogi

Adaptive server update options

After each round: evaluate on global and per-client test sets, log accuracy / macro-F1, and continue until the configured round count is reached.

A privacy-aware system needs evidence—not labels

M6 and M7 add measurable protection and efficiency experiments around the federated update pathway.

CLIENT-LEVEL DP

Clip client model deltas
Add Gaussian noise to updates
RDP accountant for ε at chosen δ
State sampling and adjacency assumptions

SECURITY & ROBUSTNESS

Membership-inference classifier
Coordinate median / trimmed mean
Synthetic faulty-client update test
Flower SecAgg+ project scaffold

COMMUNICATION EFFICIENCY

Measure parameter bytes / round
Compare ResNet50 vs EfficientNet-B0
INT8 quantisation and top-k sparsity
Local-epoch / round trade-off

IMPORTANT DISTINCTIONS

- DP is client-update level in the notebook; the accountant is only meaningful under its stated sampling and clipping assumptions.
- SecAgg+ template generation is not a verified cryptographic run. Complete setup, masked aggregation, dropout testing and log review before claiming secure aggregation.

Evaluate quality, fairness, explanation and generalisation

M8 consolidates model-level and client-level evidence instead of relying on accuracy alone.

PREDICTIVE PERFORMANCE

Accuracy • macro precision / recall / F1
Confusion matrix • one-vs-rest ROC-AUC

CLIENT FAIRNESS

Per-hospital accuracy
Worst-client performance
Compare IID and Dirichlet label skew

EXPLAINABILITY

Grad-CAM target layers for selected CNN
Save image overlays in a gallery
Review whether heatmaps are plausible

OPTIONAL BRATS EXTERNAL TEST

Authorised BraTS 2021 NIfTI → T1ce / FLAIR slice extraction → glioma-only evaluation. Not a four-class external test.

REPEATED-SEED STATISTICS

Three seeds (11, 22, 33) for selected methods and Dirichlet α=0.5; mean ± SD and paired tests on matched seeds.

Do not present smoke-test outputs, empty placeholders or unrun optional experiments as final research findings.

Turn the trained model into a reproducible project bundle

M9 generates model assets, inference interfaces, container templates and an evidence package.

MODEL EXPORT

Checkpoint + class metadata

FASTAPI

Image upload + prediction endpoint

STREAMLIT

Interactive image demo + metrics

DOCKER

Requirements + Dockerfile + Compose

FLOWER APP

Separate server + three client containers

SINGLE-HOST RESEARCH DEMO

Dashboard and API bundle is a local demonstration. It does not establish a secure multi-institution production deployment.

MULTI-SITE DEPLOYMENT PATH

Each institution needs its own controlled client environment, secured networking, access controls and verified Flower workflow.

Final outputs: zipped bundle • manifest • report template • plots / CSV / JSON • checkpoints • Grad-CAM gallery • audit table.

A complete framework—with explicit execution gates

The notebook provides the end-to-end code path; completion claims must be supported by executed artifacts and independent checks.

WHAT THE NOTEBOOK PROVIDES

Three-source data pipeline and client simulation
CNN baselines + federated training variants
DP / MIA / robustness / communication utilities
Grad-CAM, evaluation and deployment scaffolds
Report, manifest and evidence audit

WHAT MUST STILL BE VERIFIED

Kaggle permissions, dataset structure and licence
BraTS access + case-disjoint external evaluation
Actual Flower SecAgg+ run and client-dropout test
Docker build, API, dashboard and Flower containers
Three-seed experiment outputs and statistical tests

RESEARCH CONTRIBUTION

A reproducible experimental framework for studying how client heterogeneity, aggregation strategy, update privacy, communication compression and explainability interact in multi-source brain MRI classification.

Limitations: simulated clients • no guaranteed patient IDs • pHash is image-level only • BraTS is glioma-only • not for clinical use.